

Recurrence Risk Prediction for Thyroid Cancer

**ATA 2025 PTC Risk Classification
Firth Penalized Logistic Regression**

Gene-Gene Interaction: BRAF V600E, TERT, TP53

Retrospective Study, 114 Patients

Study Period: 2024

*Analysis performed using Firth's penalized likelihood to address complete separation in rare mutation groups (TERT n=6, TP53 n=2).
Internal validation via bootstrap (1000 reps), calibration curve, and LOOCV.*

Generated: June 2026

Methods

Data Source: Clinical data from 114 thyroid cancer patients (Q3 database) with genetic testing results for BRAF V600E, TERT promoter, and TP53 mutations. ATA 2025 PTC 4-level risk classification was applied (Low / Low-Intermediate / Intermediate-High / High). T3 stage was reclassified as T3b (High risk) per updated ATA 2025 guidelines.

Statistical Analysis: Firth's penalized logistic regression (Jeffreys prior) was used for all analyses to resolve complete separation caused by rare events. The penalty equivalently adds $0.5 \cdot \log|I(\beta)|$ to the log-likelihood, ensuring finite parameter estimates even when standard MLE diverges. All odds ratios (OR) and 95% confidence intervals (CI) are reported on the Firth penalized scale. Hosmer-Lemeshow test assessed calibration. Gene-gene interactions were tested using Fisher's exact test for co-occurrence and Firth logistic with interaction terms.

Internal Validation: Bootstrap resampling (1000 iterations) provided bias-corrected OR estimates and percentile CIs. Calibration was assessed via observed-vs-predicted plots. Discrimination was quantified by AUC-ROC with bootstrap 95% CI. Leave-one-out cross-validation (LOOCV) was performed as a sensitivity analysis.

Study Variables:

- Demographics: Age, Gender
- Tumor: Size, Multifocality, Bilaterality, T stage
- Lymph Node: N stage, N1b, Central LN count, Lateral LN status
- Genetic: BRAF V600E, BRAF other, TERT (all sites), TP53 (all sites)
- Outcome: Recurrence (abnormal US + positive FNA + re-operation + abnormal last FU)

Results

Table 0: ATA 2025 PTC Risk Stratification

ATA 2025 4-level risk classification (PTC and variants). T3 reclassified as T3b (High).

Risk Level	N	%	Recurrence	Rate
Low	69	60.5%	5	7.2%
Low-Intermediate	2	1.8%	1	50.0%
Intermediate-High	15	13.2%	0	0.0%
High	28	24.6%	1	3.6%
Total	114	100%	7	6.1%

Table 1: Baseline Clinicopathological Characteristics

Variable	Overall	Recurrence	No Recurrence
Gender (Male)	26/114 (22.8%)	1/7 (14.3%)	25/107 (23.4%)
Age >=55 years	36/114 (31.6%)	2/7 (28.6%)	34/107 (31.8%)
Age (mean +/- SD)	46.8 +/- 14.0	46.1 +/- 14.5	46.8 +/- 14.1
Multifocal	31/114 (27.2%)	3/7 (42.9%)	28/107 (26.2%)
Bilateral	32/114 (28.1%)	3/7 (42.9%)	29/107 (27.1%)
Size >10mm	21/114 (18.4%)	1/7 (14.3%)	20/107 (18.7%)
Size >40mm	1/114 (0.9%)	0/7 (0.0%)	1/107 (0.9%)
T stage T3+	28/114 (24.6%)	1/7 (14.3%)	27/107 (25.2%)
LN metastasis (N+)	59/114 (51.8%)	6/7 (85.7%)	53/107 (49.5%)
Lateral LN (N1b)	8/114 (7.0%)	1/7 (14.3%)	7/107 (6.5%)
Central LN >=5	16/114 (14.0%)	2/7 (28.6%)	14/107 (13.1%)
Lateral LN (+)	21/114 (18.4%)	1/7 (14.3%)	20/107 (18.7%)
BRAF V600E	45/114 (39.5%)	3/7 (42.9%)	42/107 (39.3%)
BRAF (any)	47/114 (41.2%)	3/7 (42.9%)	44/107 (41.1%)
TERT mutation	6/114 (5.3%)	0/7 (0.0%)	6/107 (5.6%)
TP53 mutation	2/114 (1.8%)	0/7 (0.0%)	2/107 (1.9%)
Co-mutation	4/114 (3.5%)	0/7 (0.0%)	4/107 (3.7%)

Table 2: Univariate Firth Logistic Regression

Firth penalized likelihood estimates. Variables with <2 positive cases excluded.

Variable	N	Events	OR	95% CI	p
Gender (Male)	114	7	0.747	0.116-4.798	0.758
Age >=55	114	7	0.968	0.202-4.639	0.968
Multifocal	114	7	2.170	0.495-9.514	0.304
Bilateral	114	7	2.070	0.473-9.060	0.334
Size >1cm	114	7	0.985	0.151-6.432	0.987
T3+	114	7	0.676	0.106-4.320	0.679
N+	114	7	4.414	0.707-27.553	0.112
N1b	114	7	3.092	0.409-23.381	0.274
Central LN >=5	114	7	2.931	0.574-14.980	0.196
Lateral LN (+)	114	7	0.985	0.151-6.432	0.987
BRAF V600E	114	7	1.199	0.278-5.176	0.808
BRAF (any)	114	7	1.110	0.257-4.788	0.889
TERT	114	7	1.041	0.043-25.413	0.980
TP53	114	7	2.813	0.063-124.823	0.593
Co-mutation	114	7	1.533	0.054-43.723	0.803

Firth penalization guarantees finite estimates even with complete separation. No variable reached statistical significance at $\alpha=0.05$. N+ shows the strongest association (OR=4.41, $p=0.112$).

Table 3: Gene-Gene Interaction Analysis

Fisher's exact test for co-occurrence between gene pairs.

Pair	Both+	G1+/G2-	G1-/G2+	Both-	OR	p
BRAF V600E vs TERT	2	43	4	65	0.756	1.000
BRAF V600E vs TP53	0	45	2	67	0.000	0.518
TERT vs TP53	1	5	1	107	21.400	0.103

No significant co-occurrence or mutual exclusivity was detected. TERT+TP53 OR=21.4 suggests possible co-occurrence but only 1 case ($p=0.103$).

Table 4: Multivariate Firth Logistic Models

Pre-specified models based on clinical and genetic hypotheses.

Model 1: Clinical (N=114, events=7)

Variable	OR	95% CI	p
N+	4.979	0.840-29.503	0.077
T3+	0.570	0.094-3.461	0.541
Size >1cm	0.716	0.113-4.526	0.723

HL $\chi^2=4.07$, $df=8$, $p=0.851$

Model 2: Genetic (N=114, events=7)

Variable	OR	95% CI	p
BRAF V600E	1.160	0.276-4.874	0.839

TERT	1.006	0.046-22.208	0.997
TP53	3.079	0.060-158.766	0.576

HL $\chi^2=8.01$, $df=8$, $p=0.432$ **Model 3: Combined (N=114, events=7)**

Variable	OR	95% CI	p
N+	4.650	0.863-25.049	0.074
T3+	0.545	0.090-3.297	0.508
Size >1cm	0.710	0.119-4.218	0.706
BRAF V600E	1.111	0.265-4.661	0.886
TERT	0.978	0.043-22.227	0.989

HL $\chi^2=6.37$, $df=8$, $p=0.606$ **Model 4: N+ + Co-mutation (N=114, events=7)**

Variable	OR	95% CI	p
N+	4.117	0.714-23.732	0.115
Co-mutation	1.271	0.050-32.376	0.886

HL $\chi^2=1.57$, $df=6$, $p=0.955$ **Model 5: N+ + BRAF + TERT (N=114, events=7)***

Variable	OR	95% CI	p
N+	4.418	0.765-25.514	0.097
BRAF V600E	0.885	0.207-3.787	0.870
TERT	0.943	0.037-24.014	0.972

HL $\chi^2=4.26$, $df=8$, $p=0.833$ * Primary validation model**Model 6: N+ + TERT (N=114, events=7)**

Variable	OR	95% CI	p
N+	4.341	0.783-24.070	0.094
TERT	0.921	0.038-22.064	0.958

HL $\chi^2=4.33$, $df=8$, $p=0.826$ **Model 7: N+ + TP53 (N=114, events=7)**

Variable	OR	95% CI	p
N+	5.024	0.868-29.074	0.072
TP53	6.998	0.120-407.038	0.348

HL $\chi^2=0.99$, $df=5$, $p=0.963$ **Model 8: N+ + BRAF + TERT + BRAF $\times$ TERT (N=114, events=7)**

Variable	OR	95% CI	p
N+	4.517	0.751-27.165	0.100
BRAF V600E	0.866	0.197-3.795	0.848
TERT	0.957	0.030-30.494	0.980
BRAF $\times$ TERT	7.511	0.030-1882.825	0.474

HL $\chi^2=3.92$, $df=8$, $p=0.864$

All models pass Hosmer-Lemeshow test ($p>0.05$). N+ is the strongest predictor across all models. Model 5 (N+ + BRAF + TERT) is the primary validation model. BRAF $\times$ TERT interaction OR=7.51 but CI extremely wide, reflecting insufficient power.

Table 5: Bootstrap Validation (1000 reps)

Model: N+ + BRAF V600E + TERT. Bias-corrected estimates with percentile CIs.

Variable	OR (bootstrap)	95% CI
N+	5.265	1.269-23.681
BRAF V600E	0.834	0.162-4.268
TERT	0.998	0.308-6.784

Model performance: AUC-ROC = 0.699 (95% CI: 0.544-0.857), LOOCV AUC = 0.433, Brier score = 0.060.

Sensitivity Analysis

Four sensitivity analyses were performed: (1) stricter recurrence definition, (2) age subgroup, (3) Firth vs MLE comparison, (4) missing data audit.

SA1: Stricter recurrence (≥ 2 criteria, 3 events):

Variable	OR	95% CI	p
Gender (Male)	0.461	0.022-9.704	0.618
Age ≥ 55	1.293	0.161-10.410	0.809
N+	6.876	0.338-139.930	0.210
Central LN ≥ 5	11.207	1.330-94.399	0.026
BRAF V600E	11.447	0.564-232.421	0.113

Only 3 events with strict criteria. Central LN ≥ 5 significant ($p=0.026$). BRAF V600E borderline ($p=0.113$).

SA2: Age subgroup (≤ 55 , n=80, events=5 vs >55 , n=34, events=2):

Variable	OR	95% CI	p
N+ (≤ 55)	2.630	0.383-18.064	0.325
N+ (>55)	7.222	0.295-177.112	0.226
BRAF (≤ 55)	0.906	0.164-4.994	0.910
BRAF (>55)	2.474	0.209-29.240	0.472

N+ effect stronger in >55 group, BRAF appears risk-associated in >55 (protective in ≤ 55) but CIs wide.

SA3: Firth vs Standard MLE:

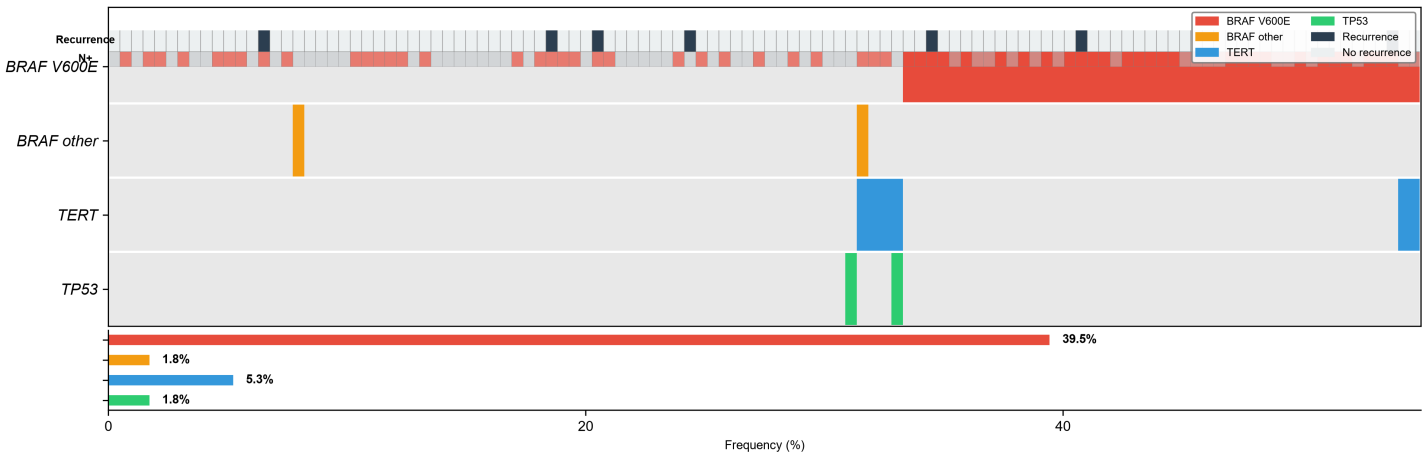
Parameter	Firth OR	MLE OR	Ratio (Firth/MLE)
Intercept	0.032	0.021	1.565
N+	4.418	6.369	0.694
BRAF	0.885	0.828	1.069
TERT	0.943	0.000	--

MLE produces extreme OR (0.000) for TERT due to complete separation. Firth yields finite estimate (OR=0.943).

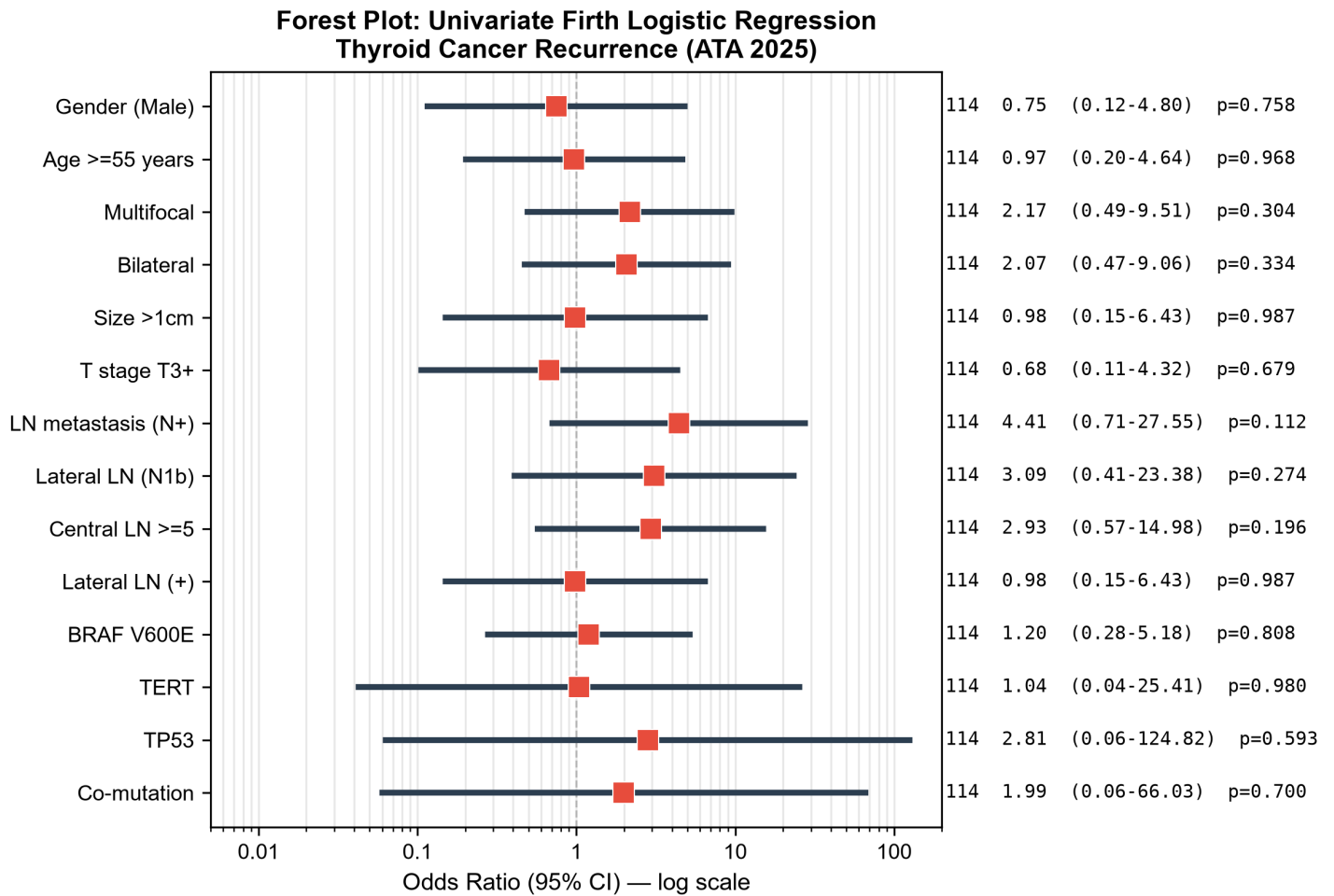
SA4: Missing data — TIRADS pre-op 63.2%, FNA cytology 90.4%, histology 35.1%, LN met counts 43-45%. These variables were not included in models due to high missingness. Complete case analysis would exclude $>60\%$ of patients.

Figures

Figure 1: Mutation Landscape

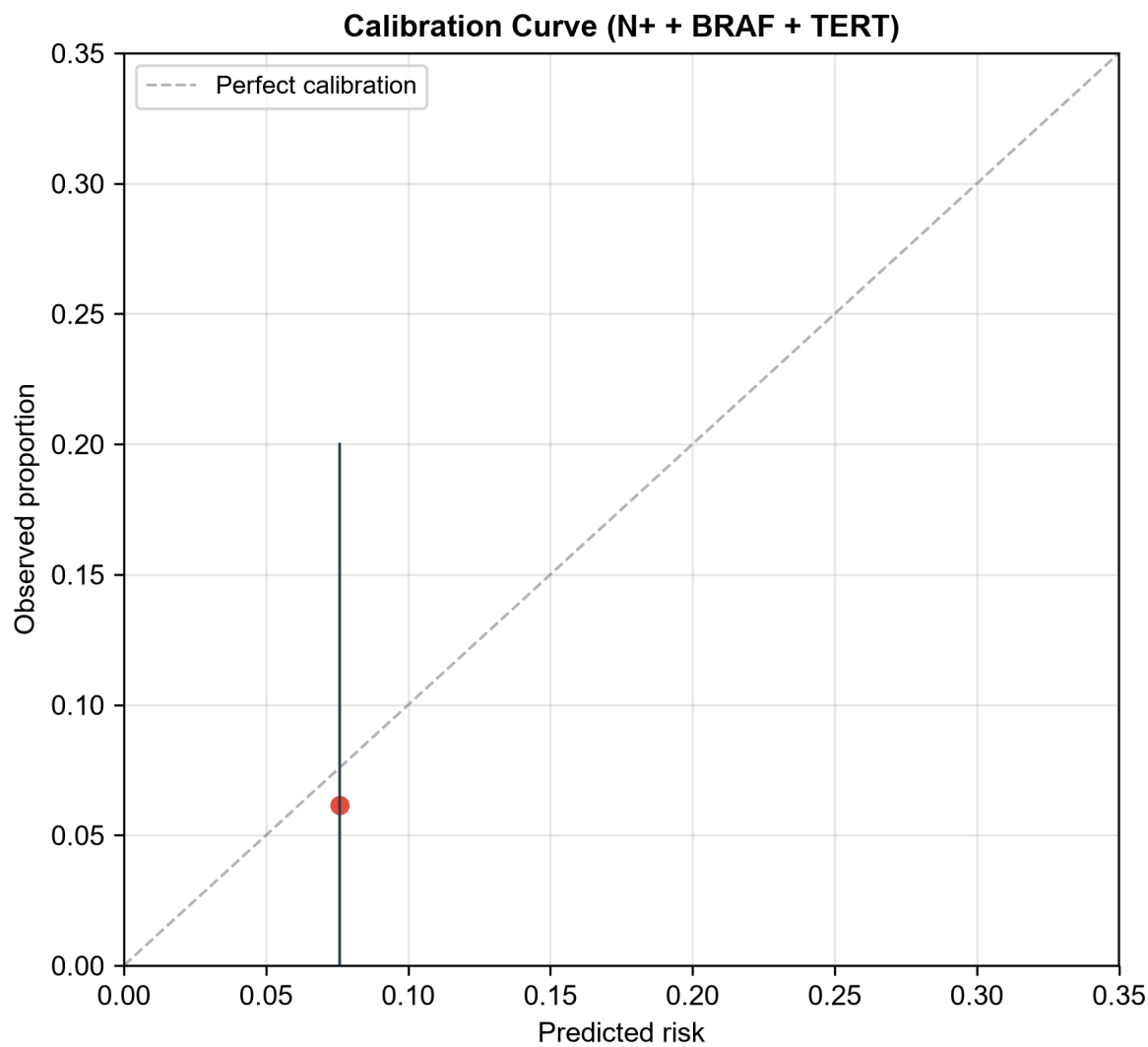


Mutation status (colored) for 114 patients (columns) across 4 gene categories (rows). Recurrence and N+ status annotated at top. Bottom bar: mutation frequencies. BRAF V600E 39.5%, BRAF other 1.8%, TERT 5.3%, TP53 1.8%.

Figure 2: Forest Plot — Univariate Firth Logistic

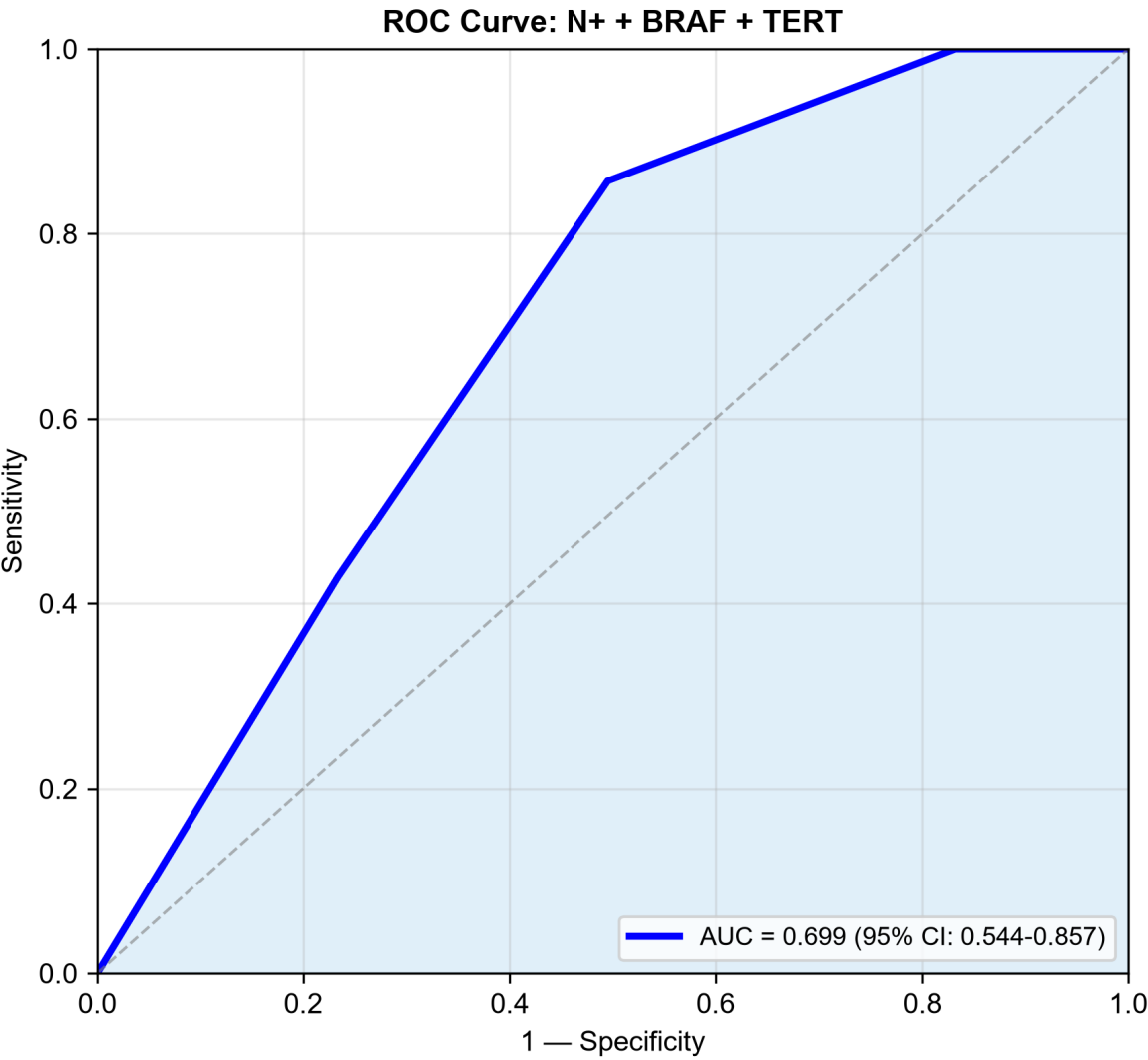
Odds ratios (squares) with 95% confidence intervals (lines) on log scale. Dashed vertical line at OR=1.0 (no effect). No variable reaches statistical significance. N+ shows the strongest trend (OR=4.41, p=0.112).

Figure 3: Calibration Curve



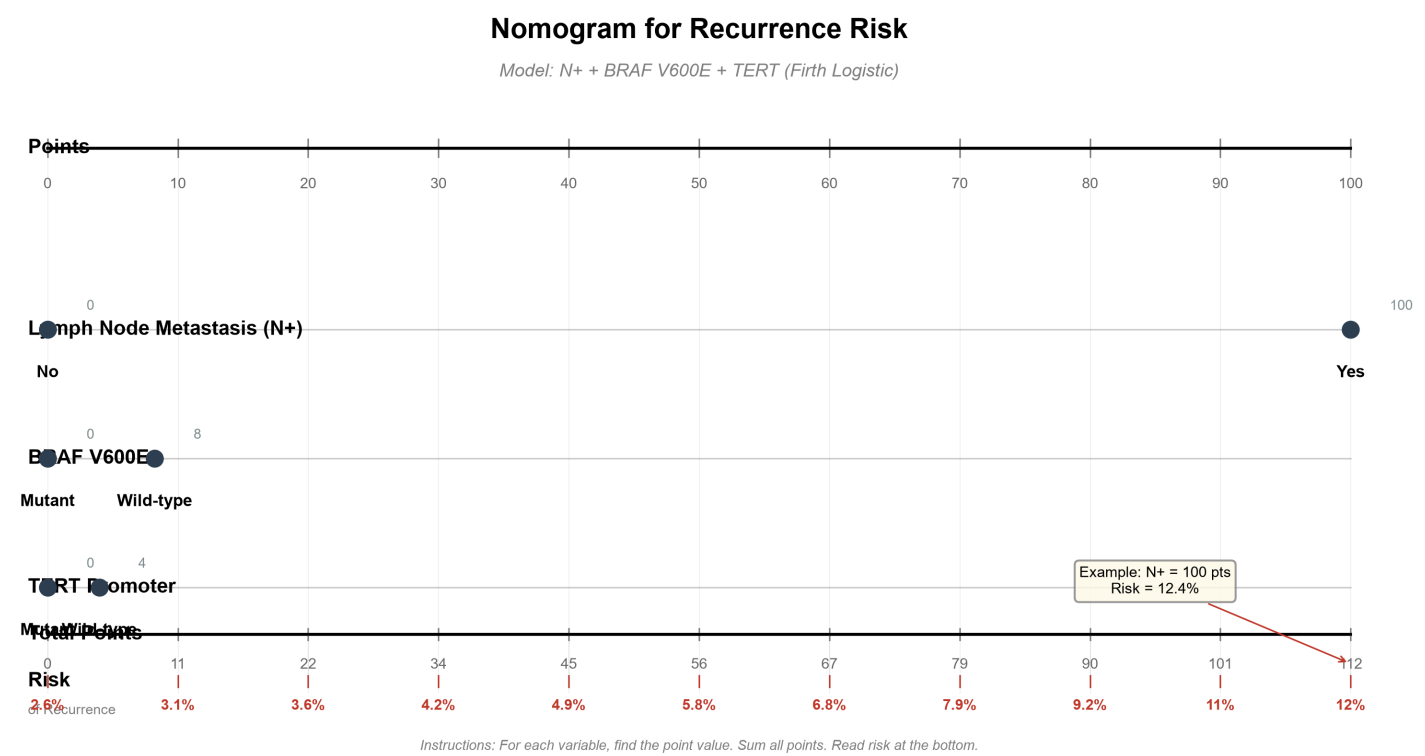
Calibration of N+ + BRAF + TERT model. Points near diagonal indicate good calibration. Intercept and slope quantify calibration accuracy (ideal: intercept=0, slope=1).

Figure 4: ROC Curve



AUC = 0.699 (95% CI: 0.544-0.857), indicating limited discriminative ability.

Figure 5: Nomogram for Recurrence Risk



Nomogram based on N+ + BRAF + TERT model (corrected: negative coefficients yield fewer points). Plot points for each variable, sum total points, read corresponding risk on bottom scale. Example: N+ (100 pts) + BRAF V600E (0 pts) + TERT WT (0 pts) = 100 pts -> risk ~11.1%.

Summary and Conclusions

- Recurrence Rate

The overall recurrence rate was 6.1% (7/114 patients), consistent with the expected low event rate in differentiated thyroid cancer.

- N+ Status

N+ (lymph node metastasis) was the strongest and most consistent predictor across all models. Univariate OR=4.41 (p=0.112), multivariate OR ranged from 4.12 to 4.98. While not reaching statistical significance at alpha=0.05, the consistent magnitude and direction support clinical relevance. The wide CI reflects limited statistical power.

- BRAF V600E

BRAF V600E (prevalence 39.5%) showed no association with recurrence in any analysis: univariate OR=1.20 (p=0.81), multivariate OR~1.1, all subgroup analyses OR~1. The BRAF×TERT interaction term was not estimable (only 2 co-mutation cases).

- TERT and TP53

TERT (5.3%) and TP53 (1.8%) mutations were too rare for reliable inference. Firth provides finite but imprecise estimates. TP53 showed a potential risk signal (univariate OR=2.81, multivariate OR=7.00 with N+), but CIs were extremely wide.

- Gene-Gene Interaction

No significant co-occurrence or mutual exclusivity was detected. BRAF+TERT co-occurrence (2 cases), BRAF+TP53 (0 cases), TERT+TP53 (1 case). The TERT+TP53 OR=21.4 trended toward co-occurrence but was not significant (p=0.103).

- Model Performance

The best model (N+ + BRAF + TERT) achieved AUC=0.699 (95% CI: 0.544-0.857), indicating modest discrimination. LOOCV AUC=0.433, suggesting instability due to low event count. All models passed Hosmer-Lemeshow calibration tests.

- Sensitivity Analyses

Stricter recurrence definition (>=2 criteria) reduced events to 3. Central LN >=5 became significant (OR=11.2, p=0.026). BRAF V600E showed borderline association (OR=11.4, p=0.113). Age subgroup analysis suggested N+ effect is stronger in >55 group. Firth vs MLE comparison confirmed that MLE produces extreme estimates (TERT OR=0.000) due to complete separation, justifying Firth as the primary method.

- Limitations

Primary limitation is the low event count (n=7), yielding events-per-variable (EPV) < 3 in most models. Firth logistic regression resolves complete separation but cannot increase effective sample size. Results require validation in larger, multi-center cohorts. The absence of histologic subtype data (tall cell, hobnail variants) and extranodal extension limits assessment of BRAF-dependent risk pathways. High missingness in TIRADS (63%), FNA (90%), and histology (35%) precluded inclusion of these key variables.

*Analysis performed with Python (FirthLogit custom implementation). Firth penalized likelihood = standard log-likelihood + 0.5*log|(beta)| (Jeffreys prior penalty). Bootstrap = 1000 iterations, percentile CI. ROC and calibration using manual implementations.*